

Naor–Young and the Heisenberg L_1 Compression Exponent

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Status

Literature-implied answer, scoped to the main logarithmic exponent. This packet is not a new proof. It records that later work of Naor–Young identifies the exponent-level answer to the compression-rate question raised by Cheeger–Kleiner–Naor, while leaving the exact lower-order compression profile open.

Source Signal

The source paper is J. Cheeger, B. Kleiner, and A. Naor, *Compression bounds for Lipschitz maps from the Heisenberg group to L_1* , arXiv:0910.2026 [1]. Theorem 1.1 (*Quantitative central collapse*) gives a universal $\delta > 0$ such that every 1-Lipschitz L_1 -valued map on a Heisenberg ball has central pairs with collapse factor at most

$$\frac{1}{(\log(1/\varepsilon))^\delta}.$$

Immediately after the theorem, the authors record that a crude estimate from their proof is $\delta = 2^{-60}$, and that obtaining the best possible δ remains an open question.

In the compression-rate subsection, Corollary 1.6 states that for every 1-Lipschitz map $f : \mathbb{H}(\mathbb{Z}) \rightarrow L_1$, with respect to a word metric,

$$\omega_f(t) \lesssim \frac{t}{(1 + \log t)^\delta} \quad (t \geq 1),$$

where ω_f is the compression rate. The following remark asks for the correct exponent in this Heisenberg-to- L_1 compression gap.

Supporting Theorem

The supporting paper is A. Naor and R. Young, *Foliated corona decompositions*, arXiv:2004.12522 [2]. In Theorem 1.10, item (5), the subgroup H used in their construction is a copy of the three-dimensional integer Heisenberg group $\mathbb{H}_{\mathbb{Z}}$. They construct a Lipschitz map $f : \mathbb{H}_{\mathbb{Z}} \rightarrow \ell_1$ whose compression satisfies

$$\omega_f(s) \gtrsim \frac{s}{\sqrt[4]{\log s} (\log \log s)^2} \quad (s \geq 3).$$

The same paper’s Remark 23 states that the lower-order factor cannot simply be deleted: there is no Lipschitz map $f : \mathbb{H}_{\mathbb{Z}} \rightarrow \ell_1$, for the word metric, whose compression rate satisfies

$$\omega_f(s) \gtrsim \frac{s}{\sqrt[4]{\log s}} \quad (s \geq 2).$$

Identification

Together, these statements pin down the principal logarithmic exponent in the $\mathbb{H}_{\mathbb{Z}} \rightarrow L_1$ compression problem as $1/4$, modulo lower-order factors. The lower-bound construction gives compression at

$$s/((\log s)^{1/4}(\log \log s)^2),$$

and the endpoint obstruction rules out a uniform compression lower bound of order $s/(\log s)^{1/4}$ with no unbounded lower-order loss. Equivalently, for every $\eta > 0$, the lower-bound construction gives $\omega_f(s) \gtrsim_{\eta} s/(\log s)^{1/4+\eta}$ eventually, while the exact $1/4$ endpoint cannot be attained without some lower-order loss.

This is an agent-identified implication of the later Naor–Young theorem and remark rather than a statement that the supporting authors explicitly framed as an answer to this exact sentence of arXiv:0910.2026.

Scope Limitations

This packet does not claim that every form of the central-collapse theorem in [1] has an optimal exponent proved in the same formulation. It records the compression-rate consequence for the discrete Heisenberg group. It also does not give a full characterization of all possible lower-order compression rates: Naor–Young explicitly state after Remark 23 that obtaining such a characterization would require additional work.

Local run indexes were searched for arXiv:0910.2026, the source title, Heisenberg L_1 compression, central collapse, Naor–Young, arXiv:2004.12522, and $\sqrt[4]{\log}$. Nearby packets already record the finite-ball distortion consequence $c_1(\mathcal{B}_n) \asymp (\log n)^{1/4}$, but no exact prior packet for this compression-rate signal was found.

Human Review Recommendation

Accept as a scoped literature-implied answer: the main logarithmic compression exponent is $1/4$ up to necessary lower-order loss. Do not count this as an original solve, and do not mark the full lower-order compression-profile problem closed.

References

- [1] J. Cheeger, B. Kleiner, and A. Naor. *Compression bounds for Lipschitz maps from the Heisenberg group to L_1* . arXiv:0910.2026, 2009.
- [2] A. Naor and R. Young. *Foliated corona decompositions*. arXiv:2004.12522, 2020.